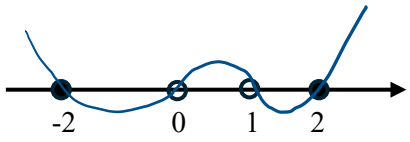
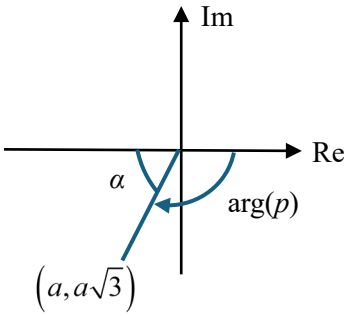
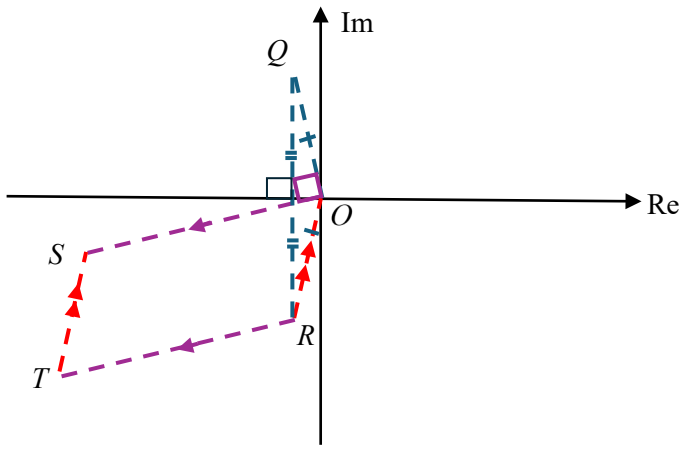
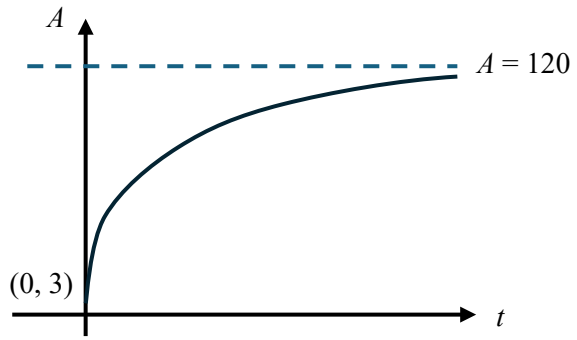
2026 VJC H2 Math Prelim Paper 1 (Solution)

1	Solution
(a)	$\dfrac{x+4}{x} \le \dfrac{3}{x-1}$ $\dfrac{(x+4)(x-1)-3x}{x(x-1)} \le 0$ $\dfrac{x^2-4}{x(x-1)} \le 0$ $\dfrac{(x+2)(x-2)}{x(x-1)} \le 0$  $-2 \le x < 0$ or $1 < x \le 2$
(b)	Replace x with x^2 $-2 \le x^2 < 0$ or $1 < x^2 \le 2$ (NA since $x^2 \ge 0$) $-\sqrt{2} \le x < -1$ or $1 < x \le \sqrt{2}$

2	Solution
(a)	$\dfrac{4a\mathrm{i}}{\sqrt{3}+\mathrm{i}} = \dfrac{4a\mathrm{i}}{\sqrt{3}+\mathrm{i}} \times \dfrac{\sqrt{3}-\mathrm{i}}{\sqrt{3}-\mathrm{i}}$ $= \dfrac{4a\mathrm{i}\left(\sqrt{3}-\mathrm{i}\right)}{3+1^2}$ $= a\mathrm{i}\left(\sqrt{3}-\mathrm{i}\right)$ $= a + a\sqrt{3}\ \mathrm{i}$  $\lvert p\rvert = \lvert a + a\sqrt{3}\ \mathrm{i}\rvert$ $= \sqrt{a^2 + 3a^2}$ $= 2\sqrt{a^2}$ $= -2a$ (since $a < 0$) $\alpha = \tan^{-1}\left\lvert\dfrac{a\sqrt{3}}{a}\right\rvert$ $\arg(p) = -\pi + \alpha = -\pi + \tan^{-1}\sqrt{3} = -\dfrac{2\pi}{3}$

(b)



3	**Solution**
(a)	As the area covered by the plant approaches 120 hectares, the rate of growth of area covered by the plant decreases and approaches zero.
(b)	$\dfrac{\mathrm{d}A}{\mathrm{d}t} = k(120 - A)$ $\dfrac{1}{(120-A)}\dfrac{\mathrm{d}A}{\mathrm{d}t} = k$ $-\int \dfrac{-1}{120-A}\,\mathrm{d}A = \int k\,\mathrm{d}t$ $-\ln\lvert 120 - A\rvert = kt + C$ $\ln\lvert 120 - A\rvert = -kt - C$ $\lvert 120 - A\rvert = D\mathrm{e}^{-kt}$ $120 - A = B\mathrm{e}^{-kt}$ $A = 120 - B\mathrm{e}^{-kt}$ At $t = 0, A = 3$ $3 = 120 - B \Rightarrow B = 117$ At $t = 4, A = 18$ $18 = 120 - 117\mathrm{e}^{-4k}$ $k = 0.0343$ (3sf) **Alternatively,** Deduced from part (a), A cannot exceed 120, $120 - A > 0$. $-\ln(120 - A) = kt + C$ $120 - A = B\mathrm{e}^{-kt}$ $A = 120 - B\mathrm{e}^{-kt}$
(c)	

4	**Solution**
(a)	At time t, $XB = 30 - x$ and $BY = y$ $\tan\angle XCB = \dfrac{30-x}{30}$ and $\tan\angle BCY = \dfrac{y}{30}$ $\tan\theta = \tan\left(\angle XCB + \angle BCY\right)$ $= \dfrac{\tan\angle XCB + \tan\angle BCY}{1 - \tan\angle XCB\tan\angle BCY}$ $= \dfrac{\dfrac{30-x}{30} + \dfrac{y}{30}}{1 - \dfrac{30-x}{30}\left(\dfrac{y}{30}\right)}$ $= \dfrac{\dfrac{30-x+y}{30}}{\dfrac{900-(30y-xy)}{900}}$ $= \dfrac{30(30-x+y)}{900-30y+xy}$ (shown)
(b)	**Method 1** Given $y = 3x$, $\tan\theta = \dfrac{30(30+2x)}{900-90x+3x^2} = \dfrac{10(30+2x)}{300-30x+x^2}$ Differentiate with respect to x, $\sec^2\theta\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{10\left(2(300-30x+x^2)-(-30+2x)(30+2x)\right)}{(300-30x+x^2)^2}$ $= \dfrac{20(750-30x-x^2)}{(300-30x+x^2)^2}$ At stationary point, $\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = 0$. $750 - 30x - x^2 = 0$ From GC, $x = 16.225$ (5 s.f.) (as $0 \le x < 30$) $\therefore$ θ is maximum when $x = 16.2$ (3 s.f.) **Method 2** $\tan\theta = \dfrac{10(30+2x)}{300-30x+x^2}$ $\theta = \tan^{-1}\left(\dfrac{10(30+2x)}{300-30x+x^2}\right)$ $\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = 0$ $\dfrac{1}{1+\left(\dfrac{10(30+2x)}{300-30x+x^2}\right)^2} \times \dfrac{10(1500-60x-2x^2)}{(300-30x+x^2)^2} = 0$

$$\frac{20(750-30x-x^2)}{(300-30x+x^2)^2+\big(10(30+2x)\big)^2}=0$$

$$750-30x-x^2=0$$

From GC, $x=16.225$ (5 s.f.) (as $0\le x<30$)

$\therefore\ \theta$ is maximum when $x=16.2$ (3 s.f.)

Method 3

Given $y = 3x$,

$$\tan\theta=\frac{30(30+2x)}{900-90x+3x^2}=\frac{10(30+2x)}{300-30x+x^2}>0 \quad \text{for } 0\le x<30$$

Hence, to maximise θ, it is to maximise $\tan\theta$.

$$\frac{\mathrm{d}}{\mathrm{d}x}(\tan\theta)=\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{10(30+2x)}{300-30x+x^2}\right]$$

$$=\frac{10\big(2(300-30x+x^2)-(-30+2x)(30+2x)\big)}{(300-30x+x^2)^2}$$

$$=\frac{20(750-30x-x^2)}{(300-30x+x^2)^2}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(\tan\theta)=0 \quad\Rightarrow\quad 750-30x-x^2=0$$

From GC, $x=16.225$ (5 s.f.) (as $0\le x<30$)

$\therefore\ \theta$ is maximum when $x=16.2$ (3 s.f.)

5	Solution
(a)	$\int(\pi-2x)\sin x\ \mathrm{d}x$ $=(\pi-2x)(-\cos x)-\int(-2)(-\cos x)\ \mathrm{d}x$ $=(2x-\pi)\cos x-2\sin x+C$ $u=\pi-2x$ and $\frac{\mathrm{d}v}{\mathrm{d}x}=\sin x$ $\frac{\mathrm{d}u}{\mathrm{d}x}=-2$ $\quad v=-\cos x$
(b)	Area bounded by 1st curve $=\int_0^{\pi}\lvert\pi-2x\rvert\sin x\ \mathrm{d}x$ $=\int_0^{\frac{\pi}{2}}(\pi-2x)\sin x\ \mathrm{d}x+\int_{\frac{\pi}{2}}^{\pi}-(\pi-2x)\sin x\ \mathrm{d}x$ $=\Big[(2x-\pi)\cos x-2\sin x\Big]_0^{\frac{\pi}{2}}-\Big[(2x-\pi)\cos x-2\sin x\Big]_{\frac{\pi}{2}}^{\pi}$ $=\big[-2-(-\pi)\big]-\big[-\pi-(-2)\big]$ $=(\pi-2)-(2-\pi)$ $=2\pi-4$

	Area bounded by 2nd curve $= -\int_{\pi}^{2\pi} \cos^2 x - 1 \, dx$ $= -\int_{\pi}^{2\pi} \frac{\cos 2x + 1}{2} - 1 \, dx$ $= -\frac{1}{2}\int_{\pi}^{2\pi} \cos 2x - 1 \, dx$ $= -\frac{1}{2}\left[\frac{1}{2}\sin 2x - x\right]_{\pi}^{2\pi}$ $= -\frac{1}{2}\left[-2\pi - (-\pi)\right]$ $= \frac{\pi}{2}$ Required Area $= 2(2\pi - 4) + \frac{\pi}{2} = \frac{9\pi}{2} - 8$

6	**Solution**
(a)	$x^2 + y^2 = 2ax + 2by - 16$ $(x-a)^2 + (y-b)^2 - a^2 - b^2 = -16$ $(x-a)^2 + (y-b)^2 = a^2 + b^2 - 16$ Centre of C : (a, b) and Radius of $C = 3$ $b = -3a + 5$ ----- (1) $a^2 + b^2 - 16 = 3^2$ $a^2 + b^2 = 25$ ----- (2) Sub (1) into (2): $a^2 + (-3a+5)^2 = 25$ $10a^2 - 30a = 0$ $10a(a-3) = 0$ $a = 3$ or $a = 0$ (reject since $a \neq 0$) From (1): $b = -4$ Equation of C : $(x-3)^2 + (y+4)^2 = 3^2 + (-4)^2 - 16$ $(x-3)^2 + (y+4)^2 = 9$ (Shown)
(b)	**Method 1:** Substitute parametric equation into circle $\left(3 + \sqrt{7t} - 3\right)^2 + (2t - 6 + 4)^2 = 9$ $7t + 4(t-1)^2 = 9$ $4t^2 - t - 5 = 0$ $(4t-5)(t+1) = 0$ $t = \frac{5}{4}$ $(\because t \geq 0)$ $\frac{dx}{dt} = \frac{\sqrt{7}}{2\sqrt{t}}$ and $\frac{dy}{dt} = 2$ $\Rightarrow \frac{dy}{dx} = \frac{4\sqrt{t}}{\sqrt{7}}$

When $t=\frac{5}{4}$, $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{4\sqrt{\frac{5}{4}}}{\sqrt{7}}=2\sqrt{\frac{5}{7}}$

(Not recommended) Method 2: Substitute cartesian equation of D into circle

$x=3+\sqrt{7t} \quad \Rightarrow t=\frac{(x-3)^2}{7}$

Curve D: $y=\frac{2(x-3)^2}{7}-6$

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{4(x-3)}{7}$$

$$(x-3)^2+\left(\frac{2(x-3)^2}{7}-6+4\right)^2=9$$

$$(x-3)^2+\left(\frac{2}{7}\right)^2\left((x-3)^2-7\right)^2=9$$

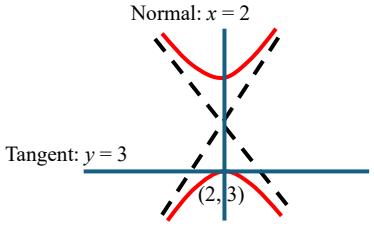
$$49(x-3)^2+4\left((x-3)^4-14(x-3)^2+49\right)=441$$

$$4(x-3)^4-7(x-3)^2-245=0$$

$$(x-3)^2=-7 \text{ (NA) or } \frac{35}{4}$$

$$x-3=\frac{\sqrt{35}}{2} \quad (\text{since } x>3)$$

At the intersection point, gradient of D: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{4(x-3)}{7}=\frac{2\sqrt{35}}{7}$

7	**Solution**
(a)	$\frac{(y-4)^2}{1^2}-\frac{(x-2)^2}{1^2}=1$ Centre of the hyperbola C is (2, 4). Turning points (vertices) of C: $(2,4\pm1)=(2,3)$ and $(2,5)$
(b)	Equation of tangent is $y=3$ Equation of normal is $x=2$ 
(c)	Differentiate $(y-4)^2-(x-2)^2=1$ w.r.t. x $2(y-4)\frac{\mathrm{d}y}{\mathrm{d}x}-2(x-2)=0$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{x-2}{y-4}$

	When $y = 6$, $2^2 - (x-2)^2 = 1$ $(x-2)^2 = 3$ $x = 2 - \sqrt{3}$ or $2+\sqrt{3}$ $\frac{dy}{dx} = \frac{-\sqrt{3}}{2}$ or $\frac{\sqrt{3}}{2}$ Equation of tangents are: $y - 6 = -\frac{\sqrt{3}}{2}\left(x - (2-\sqrt{3})\right)$ and $y - 6 = \frac{\sqrt{3}}{2}\left(x - (2+\sqrt{3})\right)$ $y = -\frac{\sqrt{3}}{2}x + \sqrt{3} + \frac{9}{2}$ $y = \frac{\sqrt{3}}{2}x - \sqrt{3} + \frac{9}{2}$

8	**Solution**
(a)	As $n \to \infty$, $\frac{1}{4n+2} \to 0$, $\therefore \sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)} \to \frac{1}{2}$, a constant. Hence $\sum_{r=1}^{\infty} \frac{1}{(2r-1)(2r+1)}$ converges and $\sum_{r=1}^{\infty} \frac{1}{(2r-1)(2r+1)} = \frac{1}{2}$.
(b)	Since $4r^2 > 4r^2 - 1$, $\frac{1}{4r^2} < \frac{1}{4r^2-1}$. $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \ldots = \sum_{r=1}^{\infty} \frac{1}{r^2}$ $= 4\sum_{r=1}^{\infty} \frac{1}{4r^2}$ $< 4\sum_{r=1}^{\infty} \frac{1}{(2r-1)(2r+1)} = 4\left(\frac{1}{2}\right) = 2$

(c)

Method 1

$$\sum_{r=10}^{N}\frac{1}{(2r+1)(2r+3)}$$

$$=\sum_{r-1=10}^{r-1=N}\frac{1}{(2r-1)(2r+1)}$$

$$=\sum_{r=11}^{N+1}\frac{1}{(2r-1)(2r+1)}$$

$$=\sum_{r=1}^{N+1}\frac{1}{(2r-1)(2r+1)}-\sum_{r=1}^{10}\frac{1}{(2r-1)(2r+1)}$$

$$=\left[\frac{1}{2}-\frac{1}{4(N+1)+2}\right]-\left[\frac{1}{2}-\frac{1}{4(10)+2}\right]$$

$$=\frac{1}{42}-\frac{1}{4N+6}$$

Method 2

$$\sum_{r=10}^{N}\frac{1}{(2r+1)(2r+3)}$$

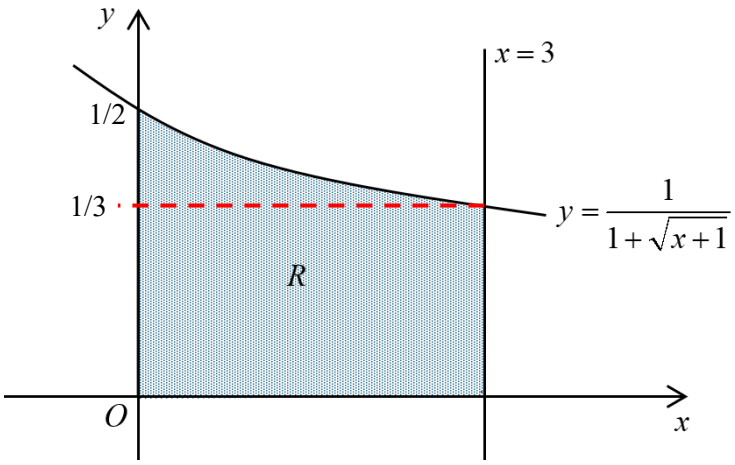
$$=\frac{1}{(21)(23)}+\frac{1}{(23)(25)}+\ldots+\frac{1}{(2N+1)(2N+3)}$$

$$=\sum_{r=11}^{N+1}\frac{1}{(2r-1)(2r+1)}$$

$$=\sum_{r=1}^{N+1}\frac{1}{(2r-1)(2r+1)}-\sum_{r=1}^{10}\frac{1}{(2r-1)(2r+1)}$$

$$=\left[\frac{1}{2}-\frac{1}{4(N+1)+2}\right]-\left[\frac{1}{2}-\frac{1}{4(10)+2}\right]$$

$$=\frac{1}{42}-\frac{1}{4N+6}$$

9	**Solution**
(a)	

$x=0,\ y=\frac{1}{1+\sqrt{1}}=\frac{1}{2}$ and $x=3,\ y=\frac{1}{1+\sqrt{4}}=\frac{1}{3}$

	Making x the subject, $$y=\frac{1}{1+\sqrt{x+1}}$$ $$y+y\sqrt{x+1}=1$$ $$x=\left(\frac{1-y}{y}\right)^2-1$$ $$\text{Volume}=\pi(3)^2\left(\frac{1}{3}\right)+\pi\int_{\frac{1}{3}}^{\frac{1}{2}}\left[\left(\frac{1-y}{y}\right)^2-1\right]^2\,\mathrm{d}y\approx 10.5$$
(b)	**Method 1 – using partial fractions** $u=\sqrt{x+1}$ When $x = 0$, $u = 1$ When $x = 3$, u 2 $$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{2\sqrt{x+1}}=\frac{1}{2u}\quad\Rightarrow\quad 2u\ \mathrm{d}u=\mathrm{d}x$$ $$\text{Volume}=\pi\int_0^3\left(\frac{1}{1+\sqrt{x+1}}\right)^2\mathrm{d}x$$ $$=\pi\int_1^2\left(\frac{1}{1+u}\right)^2(2u)\,\mathrm{d}u$$ $$=\pi\int_1^2\frac{2u}{(1+u)^2}\,\mathrm{d}u$$ $$=\pi\int_1^2\frac{2}{1+u}-\frac{2}{(1+u)^2}\,\mathrm{d}u$$ $$=\pi\left[2\ln(1+u)+2(u+1)^{-1}\right]_1^2\quad(\text{since } u>0)$$ $$=\pi\left[\left(2\ln 3+\frac{2}{3}\right)-\left(2\ln 2+\frac{2}{2}\right)\right]$$ $$=\pi\left(-\frac{1}{3}+2\ln\frac{3}{2}\right)$$ **Method 2 – using splitting of numerator** $$\text{Volume}=\pi\int_0^3\left(\frac{1}{1+\sqrt{x+1}}\right)^2\mathrm{d}x$$

$$= \pi \int_1^2 \left(\frac{1}{1+u}\right)^2 (2u)\,\mathrm{d}u$$

$$= \pi \int_1^2 \frac{2u}{u^2+2u+1}\,\mathrm{d}u$$

$$= \pi \int_1^2 \frac{2u+2}{u^2+2u+1} - \frac{2}{(u+1)^2}\,\mathrm{d}u$$

$$= \pi \left[\ln\left(u^2+2u+1\right) + 2(u+1)^{-1}\right]_1^2 \quad \text{(since } u > 0\text{)}$$

$$= \pi \left[\left(\ln 9 + \frac{2}{3}\right) - \left(\ln 4 + \frac{2}{2}\right)\right]$$

$$= \pi \left(-\frac{1}{3} + 2\ln\frac{3}{2}\right)$$

10	**Solution**
(a)	Normal of $\pi_1 = \begin{pmatrix}1\\1\\2\end{pmatrix} \times \begin{pmatrix}2\\1\\3\end{pmatrix} = \begin{pmatrix}3-2\\-(3-4)\\1-2\end{pmatrix} = \begin{pmatrix}1\\1\\-1\end{pmatrix}$ $\cos^{-1}\dfrac{\begin{pmatrix}1\\1\\-1\end{pmatrix}\bullet\begin{pmatrix}-1\\-2\\2\end{pmatrix}}{\sqrt{3}\sqrt{9}} = \cos^{-1}\dfrac{-1-2-2}{3\sqrt{3}} = 164.207°$ Required angle $= 180° - 164.207° = 15.8°$
(b)	**Method 1** Equation of π_1: $\mathbf{r}\bullet\begin{pmatrix}1\\1\\-1\end{pmatrix} = \begin{pmatrix}-4\\1\\2\end{pmatrix}\bullet\begin{pmatrix}1\\1\\-1\end{pmatrix}$ $x+y-z=-5 \quad ---(1)$ Equation of π_2: $\mathbf{r}\bullet\begin{pmatrix}-1\\-2\\2\end{pmatrix} = 10$ $-x-2y+2z=10 \quad ---(2)$ Solve (1) and (2) using GC: $\mathbf{r} = \begin{pmatrix}0\\-5+z\\z\end{pmatrix}$ $l_1 : \mathbf{r} = \begin{pmatrix}0\\-5\\0\end{pmatrix} + \alpha\begin{pmatrix}0\\1\\1\end{pmatrix},\ \alpha \in \mathbb{R}$

Method 2

$$\left[\begin{pmatrix}-4\\1\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}+\mu\begin{pmatrix}2\\1\\3\end{pmatrix}\right]\bullet\begin{pmatrix}-1\\-2\\2\end{pmatrix}=10$$

$$(4-2+4)+\lambda(-1-2+4)+\mu(-2-2+6)=10$$

$$\lambda+2\mu=4$$

$$\lambda=4-2\mu$$

Eq of line: $\mathbf{r}=\begin{pmatrix}-4\\1\\2\end{pmatrix}+(4-2\mu)\begin{pmatrix}1\\1\\2\end{pmatrix}+\mu\begin{pmatrix}2\\1\\3\end{pmatrix}$

$$\mathbf{r}=\begin{pmatrix}0\\5\\10\end{pmatrix}+\alpha\begin{pmatrix}0\\-1\\-1\end{pmatrix},\text{ where }\alpha\in\mathbb{R}$$

(c)

Method 1

$$\overrightarrow{BA}=\begin{pmatrix}4\\-2\\3\end{pmatrix}-\begin{pmatrix}0\\-5\\0\end{pmatrix}=\begin{pmatrix}4\\3\\3\end{pmatrix}$$

$$\overrightarrow{BF}=\left(\frac{\begin{pmatrix}4\\3\\3\end{pmatrix}\cdot\begin{pmatrix}0\\1\\1\end{pmatrix}}{\sqrt{2}}\right)\frac{\begin{pmatrix}0\\1\\1\end{pmatrix}}{\sqrt{2}}$$

$$=\frac{6}{2}\begin{pmatrix}0\\1\\1\end{pmatrix}=\begin{pmatrix}0\\3\\3\end{pmatrix}$$

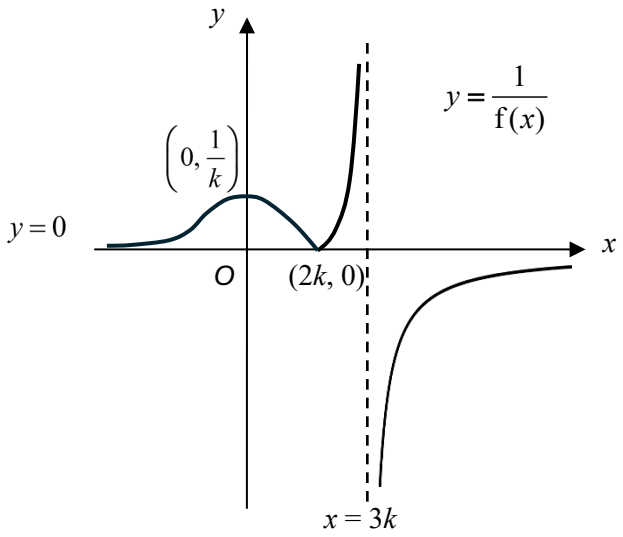
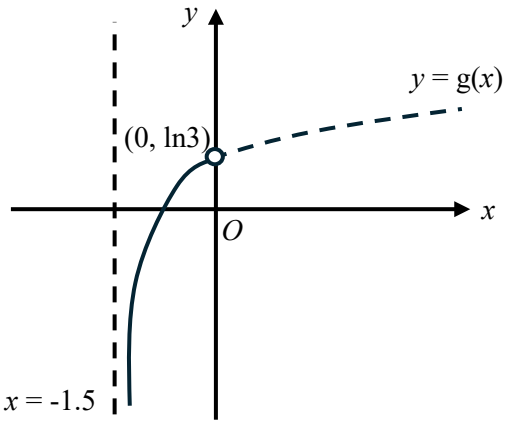
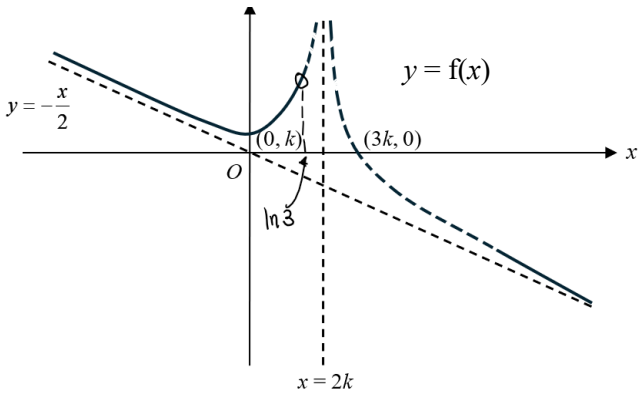
$$\overrightarrow{OF}=\overrightarrow{OB}+\overrightarrow{BF}=\begin{pmatrix}0\\-5\\0\end{pmatrix}+\begin{pmatrix}0\\3\\3\end{pmatrix}=\begin{pmatrix}0\\-2\\3\end{pmatrix}$$

Method 2

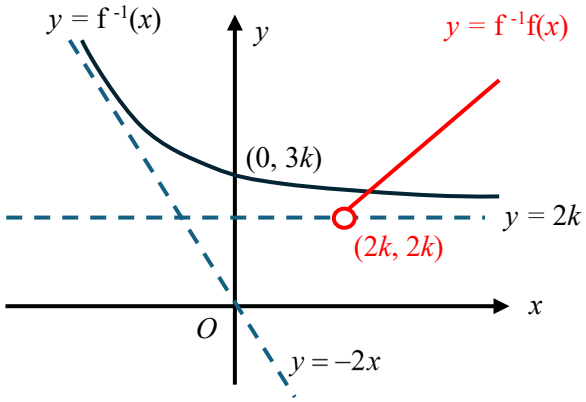
Since F is on l_1, $\overrightarrow{OF}=\begin{pmatrix}0\\-5\\0\end{pmatrix}+\alpha\begin{pmatrix}0\\1\\1\end{pmatrix}$

$$\overrightarrow{AF}=\overrightarrow{OF}-\overrightarrow{OA}=\begin{pmatrix}0\\-5\\0\end{pmatrix}+\alpha\begin{pmatrix}0\\1\\1\end{pmatrix}-\begin{pmatrix}4\\-2\\3\end{pmatrix}=\begin{pmatrix}-4\\-3+\alpha\\-3+\alpha\end{pmatrix}$$

	Since $\overrightarrow{AF} \perp \overrightarrow{BF}$, $\begin{pmatrix} -4 \\ -3+\alpha \\ -3+\alpha \end{pmatrix} \bullet \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0$ $-3+\alpha-3+\alpha = 0$ $\alpha = 3$ $\overrightarrow{OF} = \begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$
(d)	Equation of l_2: $x = m,\ y+1 = z-5$ i.e. $\underset{\sim}{r} = \begin{pmatrix} m \\ -1 \\ 5 \end{pmatrix} + \mu\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ Note that l_2 is parallel to l_1. $\Rightarrow$ l_2 is parallel to π_2 Since given π_2 and l_2 have no point in common $\Rightarrow$ $(m,\ -1,\ 5)$ is not a point on π_2. $\begin{pmatrix} m \\ -1 \\ 5 \end{pmatrix} \bullet \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} \neq 10$ $12-m \neq 10$ $m \neq 2$
(e)	Shortest distance from $(m,\ -1,\ 5)$ to $\pi_2 = 3$ $\left\| \left[\begin{pmatrix} m \\ -1 \\ 5 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \right] \bullet \frac{\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}}{\sqrt{9}} \right\| = 3$ $\lvert -m+2 \rvert = 9$ $-m+2 = 9$ or -9 $m = -7$ (NA) or 11

11	Solution
(a)	
(b)	 Range of g = $(-\infty, \ln 3)$  Range of fg = $[k, \infty)$

(c)



(d)

$$\mathrm{hh}(x) = \frac{a - \left(\frac{a-x}{1+x}\right)}{1 + \left(\frac{a-x}{1+x}\right)}$$

$$= \frac{a(1+x) - (a-x)}{(1+x) + (a-x)}$$

$$= \frac{ax + x}{1 + a}$$

$$= \frac{(1+a)x}{1+a}$$

$$= x$$

$$\mathrm{h}^3(x) = x$$

$$\frac{a-x}{1+x} = x$$

$$a - x = x + x^2$$

$$x^2 + 2x - a = 0 \quad (*)$$

Since (*) has **2 real** and **distinct** roots,

$\Rightarrow$ discriminant > 0

$$2^2 - 4(1)(-a) > 0$$

$$a > -1$$

12 Solution

(a)

Amount set aside on 1 December 2025

$= 50 + (24 - 1)(a)$

$= \$(50 + 23a)$

(b)

$50 + (50 + a) + \ldots + (50 + 23a) \geq 20000$

$$\frac{24}{2}(50 + 50 + 23a) \geq 20000$$

$$a \geq 68.1159$$

Least a = 68.12 (to the nearest cent)

(c)

End of Jan 2026: Amount in fund = $20000(1.002)$

End of Feb 2026: Amount in fund

$$= 1.002\left[(20\ 000)(1.002) - 180\right] = 19\ 899.72$$

The amount at the end of Feb 2026 = \$19 899.72

(d)

n	mth	Amount in fund at end of month
1	Jan 2026	(20 000)(1.002)
2	Feb 2026	[20 000(1.002) – 180] (1.002) = 20 000(1.002)^2 – 180(1.002)
3	Mar 2026	20 000(1.002)^3 – 180(1.002^2+1.002)
n		$20\ 000(1.002)^n - 180\left[\begin{array}{l}(1.002)^{n-1} + (1.002)^{n-2} + \\ \ldots + (1.002)^2 + 1.002\end{array}\right]$

Amount in fund at the end of n months

$$= 20\ 000(1.002)^n - 180\left[(1.002)^{n-1} + (1.002)^{n-2} + \ldots + (1.002)^2 + 1.002\right]$$

$$= 20\ 000(1.002)^n - 180\left[\frac{(1.002)\left[(1.002)^{n-1} - 1\right]}{1.002 - 1}\right]$$

$$= 20\ 000(1.002)^n - 90\ 000\left[(1.002)^n - 1.002\right]$$

$$= 90\ 180 - 70\ 000(1.002)^n$$

Hence, $k = 90\ 180$

(e)

Method 1

$$90\ 180 - 70\ 000(1.002)^n \leq 0$$

$$(1.002)^n \geq \frac{90\ 180}{70\ 000}$$

$$n \geq \frac{\ln\left(90\ 180 \big/ 70\ 000\right)}{\ln(1.002)}$$

$$n \geq 126.783$$

Account is exhausted after the last withdrawal on July 2036.

Method 2

n	$90\,180 - 70\,000(1.002)^n$
125	$320.66 > 180$
126	$140.94 < 180$

By the end of 126th month, she will have less than \$180. Therefore, account is exhausted on her next withdrawal in July 2036.

(f) $90\,180 - 70\,000(1.002)^{126} = 140.94$

Amount for last withdrawal = \$140.94